手寫作業

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[illegible]

11. Let C be a nonsymmetric $n \times n$ matrix. For each of the following, determine whether the given matrix must necessarily be symmetric or could possibly be nonsymmetric:

(a) $A = C + C^T$ **(b)** $B = C - C^T$

(c) $D = C^T C$ **(d)** $E = C^T C - C C^T$

(e) $F = (I + C)(I + C^T)$

(f) $G = (I + C)(I - C^T)$

When C is a symmetric matrix $\Leftrightarrow C = C^T$

(a)

$A = C + C^T$

$A^T = (C + C^T)^T = C^T + C$

$A = A^T$, so A is symmetric #

(b) $B = C - C^T$

$B^T = C^T - C$

When B is symmetric

$B = B^T$, $C - C^T = C^T - C$

$\Rightarrow 2C = 2C^T$

$\Rightarrow C = C^T$ (C is symmetric)

Say, B is possibly nonsymmetric #

(c) $D = C^T C$

$D^T = (C^T C)^T = (C)^T \cdot (C^T)^T = C^T \cdot C$

$D = D^T \Rightarrow D$ is symmetric #

(d)

$E = C^T C - C C^T$

$E^T = C^T C - (C^T)^T \cdot C^T$

$= C^T C - C \cdot C^T$

$E = E^T$, E is symmetric #

(e) $F = (I + C)(I + C^T) = I + C^T + C + C C^T$

$F^T = (I + C^T + C + C C^T)^T$

$= I^T + (C^T)^T + C^T + (C C^T)^T$

$= I + C + C^T + (C^T)^T C^T$

$= I + C + C^T + C C^T$

$F = F^T \Rightarrow F$ is symmetric #

(f) $G = (I + C)(I - C^T) = I - C^T + C - C C^T$

$G^T = I - C + C^T - C C^T$

when G is symmetrix, $G = G^T$

$I - C^T + C - C C^T = I - C + C^T - C C^T$

$\Rightarrow 2C = 2C^T \Rightarrow C = C^T$ (C is symmetric)

Say, G is possibly nonsymmetric #

14. Let U and R be $n \times n$ upper triangular matrices and set $T = UR$. Show that T is also upper triangular and that $t_{jj} = u_{jj} r_{jj}$ for $j = 1, \ldots, n$.

$$R = \begin{bmatrix} r_{11} & r_{12} & r_{13} & \cdots & r_{1n} \\ 0 & r_{22} & & & \vdots \\ 0 & & \ddots & & \vdots \\ \vdots & & & r_{n-1\,n-1} & \vdots \\ \underset{\hookrightarrow r_{n1}}{0} & \cdots & \cdots & \cdots & r_{nn} \end{bmatrix} \quad \& \quad \begin{cases} r_{ij} \neq 0, & \forall\, j \geq i \\ r_{ij} = 0, & \forall\, j < i \end{cases}$$

$$U = \begin{bmatrix} u_{11} & u_{12} & \cdot & u_{1n} \\ 0 & u_{22} & & \vdots \\ \vdots & & \ddots & \vdots \\ 0 & \cdots & & u_{nn} \end{bmatrix} \quad \& \quad \begin{cases} u_{ij} \neq 0, & \forall\, j \geq i \\ u_{ij} = 0, & \forall\, j < i \end{cases}$$

$T = RU$, $t_{ij} = \text{row}(R)_i \cdot \text{col}(U)_j$

To prove $t_{ij} = 0$, $\forall\, j < i$

For $\text{row}(R)_i$, 前 $i-1$ 個元是 0，後 $n-i-1 \neq 0$

For $\text{col}(U)_j$，前 j 個 $\neq 0$，後 $n-j$ 是 0

$$\underset{(j<i)}{\text{row}(R)_i \cdot \text{col}(U)_j} = \begin{bmatrix} \underbrace{0\ 0\ 0}_{i-1} & \underbrace{\cdots\ r_{in}}_{n-(i-1)} \end{bmatrix} \begin{bmatrix} \left.\begin{matrix} u_{1j} \\ u_{2j} \\ \vdots \end{matrix}\right\} j \\ \left.\begin{matrix} 0 \\ 0 \\ 0 \end{matrix}\right\} n-j \end{bmatrix}$$

To make sure the number of zero entry $\geq$ nonzero entry

① $i > j$, since $i, j \in \mathbb{Z}$, so $i \geq j+1 \Rightarrow i-1 \geq j$ (from beginning entry)

② $i > j \Rightarrow -j > -i \Rightarrow n-j > n-i \Rightarrow n-j \geq n-i+1$ (from ended entry)

Say, $\text{row}(R)_i \cdot \text{col}(U)_j = 0 \Rightarrow t_{ij} = 0$, $\forall\, j < i$ #

$$t_{jj} = \text{row}(R)_j \cdot \text{col}(U)_j, \quad j = 0, 1, 2 \cdots n$$

$$\text{row}(R)_j \cdot \text{col}(U)_j = [\underbrace{0, 0, \cdots 0}_{j-1}, r_{jj}, \underbrace{r_{j,j+1} \cdots r_{jn}}_{n-j}] \begin{bmatrix} \left.\begin{matrix} u_{1j} \\ \vdots \end{matrix}\right\} j-1 \\ u_{jj} \\ \left.\begin{matrix} 0 \\ \vdots \\ 0 \\ 0 \\ 0 \end{matrix}\right\} n-j \end{bmatrix} = r_{jj} \cdot u_{jj}$$

From the expansion of $\text{row}(R)_j \cdot \text{col}(U)_j$ we found their product of them is the product of entry $\text{row}(R)_{r(j,j)}$ and $\text{col}(U)_{j\,(j,j)}$, so we can easily know the $t_{jj} = r_{jj} \cdot u_{jj}$, $j = 0, 1, 2 \cdots n$ #

24. (a) Prove that if A is row equivalent to B and B is row equivalent to C, then A is row equivalent to C.

(b) Prove that any two nonsingular $n \times n$ matrices are row equivalent.

E_1 and E_2 are composed of finite sequence of elementary matrices and they are all nonsingular $\rightarrow$ invertible, so E_1 and E_2 are also nonsingular

(a) From the given conditions, $A = E_1 \cdot B$, $B = E_2 C$, E_1 and E_2 are invertible matrices (row)

By subsitution, $A = E_1 \cdot E_2 \cdot C$, set $E_{12} = E_1 \cdot E_2$, we can find E_{12}^{-1}. $E_{12}(E_{12}^{-1}) = I$

$E_1 \cdot E_2 \cdot E_{12}^{-1} = I$

$E_{12}^{-1} = E_2^{-1} E_1^{-1}$

Say $A = E_{12} \cdot C$ and E_{12} is invertible

$\Rightarrow$ A is row equivalent to C #

(b) set nonsingular matrices A, B

since they are nonsingular $\Rightarrow$ A, B are invertible

$A \cdot A^{-1} = I$, $B \cdot B^{-1} = I$

$\Rightarrow A^{-1} A = B^{-1} B$

$\Rightarrow A^{-1} A = B^{-1} B$

$\Rightarrow B = (BA^{-1}) A$, $(A = AB^{-1} B)$

From the provement of (a), we know the product of two invertible matrices are still invertible

$(AB)^{-1} = B^{-1} A^{-1}$

Say, (BA^{-1}) is invertible which refers B is equivalant to A

(A , B) #

11. Let

$$A = \begin{bmatrix} A_{11} & A_{12} \\ O & A_{22} \end{bmatrix}$$

where all four blocks are $n \times n$ matrices.

(a) If A_{11} and A_{22} are nonsingular, show that A must also be nonsingular and that A^{-1} must be of the form

$$\left[\begin{array}{c|c} A_{11}^{-1} & C \\ \hline O & A_{22}^{-1} \end{array}\right]$$

(b) Determine C.

(a)

create another big block matrix $\left[\begin{array}{c|c} I & 0 \\ \hline 0 & I \end{array}\right]$ then put two blocks matrices together

$$\longrightarrow \left[\begin{array}{c|c|c|c} A_{11} & A_{12} & I & 0 \\ \hline 0 & A_{22} & 0 & I \end{array}\right]$$

, by $\left[\begin{array}{c|c} I & 0 \\ \hline 0 & I \end{array}\right]$ we're able to record our row operation. Fortunately, entry $(2,1)$ is a zero matrix, so we can directly let $A_{22} \to I$ by Gaussian elimination only operating the rows downside

(只操作 A_{11}、A_{12} 以下的 rows 並持續往下做消去，所以 A_{11}, A_{12} 不會因為 A_{22} 做了高斯受到影響)

$$\longrightarrow \left[\begin{array}{c|c|c|c} A_{11} & A_{12} & I & 0 \\ \hline 0 & I & 0 & A_{22}^{-1} \end{array}\right]$$

Now, apply elementary matrix to eliminate A_{12}

put A_{12} to left side since we're doing row operation

$$\longrightarrow \left[\begin{array}{c|c|c|c} A_{11} & 0 & I & -A_{12}A_{22}^{-1} \\ \hline 0 & I & 0 & A_{22}^{-1} \end{array}\right] \times(-A_{12})$$

After that, we apply the same idea to upper rows, let $A_{11} \to I$

$$\longrightarrow \left[\begin{array}{c|c|c|c} I & 0 & A_{11}^{-1} & -A_{11}^{-1}A_{12}A_{22}^{-1} \\ \hline 0 & I & 0 & A_{22}^{-1} \end{array}\right]$$

It's easily to find the $A^{-1} = \left[\begin{array}{c|c} A_{11}^{-1} & -A_{11}^{-1}A_{12}A_{22}^{-1} \\ \hline 0 & A_{22}^{-1} \end{array}\right]$ which refers A is nonsingular (nonsingular $\Leftrightarrow$ invertible) as well. ※

(b) $C = -A_{11}^{-1}A_{12}A_{22}^{-1}$ ※

程式作業

5.

(a)

```
>> HW2_5a
solution x:
   -2.3627
    2.4012
   -1.8597
   -1.4182
    1.3545
    1.0226

seventh column of reduced row echelon form matrix:
   -2.3627
    2.4012
   -1.8598
   -1.4182
    1.3545
    1.0226

two vectors equal
```

```
% Explanation

% for the reason why two vectors are the same, we know that row operation
% would not influence the linear relationship of column vectors in the matrix
% so we can separate the augmented matrix to original form A and b, and we can find that A
% matrix become a identity matrix, so x vector multiply I = x, which means
% I * x = b' --> x = b'
```

#

$[A|b] \rightarrow [I|b'] \rightarrow Ix = b = x = b'$

nonsingular

(b)

```
>> HW2_5b
     1     0     4     0     0     0     0
     0     1     3     0     0     0     0
     0     0     0     1     0     0     0
     0     0     0     0     1     0     0
     0     0     0     0     0     1     0
     0     0     0     0     0     0     1

Normaly, this system would have no solutions
```

I wrote the reason in following explanation

```
% Explanation

% We change third column of A into the linear combination of fisrt and
% second columns combination of A, so the matrix must be singular and the
% solution of x has infinite solutions when vector b is in the column
% space of A. In other words, b is the linear combination of A's columns
% vectors so this system is consistent(b is ramdomly generated, so this
% posibility is very small). Otherwise, x has no solutions.(For most cases)
% In summary, this system is consistent and A is singular, so it would only
% have infinite solutions. This point can be simply found by observing
% rref([A b]) itself.
```

#

$col(A)_3 = 4 \times col(A)_1 + 3 \times col(A)_2$

(c)

```
>> HW_5c
     1     0     4     0     0     0    28
     0     1     3     0     0     0    20
     0     0     0     1     0     0     5
     0     0     0     0     1     0    -5
     0     0     0     0     0     1     2
     0     0     0     0     0     0     0

this system has infinite solutions
```

```
% Explanation

% it is clear that y exist and Ay = c, so we can easily see for Ax = c,
% this system at least must have the solution x = y, and this system Ax = b
% must be consistent
% For how many solutons this system has, we should look at matrix A. When
% A is singular, which means A has redundent column(s) and x will have
% free variable(s) and this system will have infinite solutions.
% On the contrary, when A is nonsingular, c is the only combination of
% A's columns, so we can find x has only one solution by equation
% A' * Ax = A' * c --> x = A' * c
% From the reasons above, A become singular at section (b) and this system
% has infinite solutions. To be specific, x_3 is the free variable.
```

(d)

```
>> HW2_5d
vector Aw and c:
A*w equals to c
   345   345
   321   321
   293   293
   190   190
   120   120
   157   157

residual vector c - Aw:
     0
     0
     0
     0
     0
     0
```

(e)

computed by matlab

$$R = \left[\begin{array}{cccccc|c} 1 & 0 & 4 & 0 & 0 & 0 & 0 \\ 0 & 1 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$$\Rightarrow \left[\quad '' \quad \right] \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} | \\ 0 \\ | \end{bmatrix}$$

let $x_3 = \alpha$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} -4\alpha \\ -3\alpha \\ \alpha \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$x_3 = \alpha = 1$

$$\Rightarrow \begin{bmatrix} | \\ x \\ | \end{bmatrix} = \begin{bmatrix} -4 \\ -3 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

```
>> HW2_5e
A * z :
     0
     0
     0
     0
     0
     0
```

(f)

```
>> HW2_5f
residual vector c - A*v:
     0
     0
     0
     0
     0
     0

solution v
:   -18
   -10
     3
    -9
    -7
     1

the value of x_3 is 3
```

$V = w + 3z$

we know that w is the solution of system, $Ax = c$

and z is the solutions of $Ax = 0$

$Aw = c$, $Az = 0$

insert v into $Ax \Rightarrow Av = A(w+3z)$

$$\Rightarrow Av = \overset{c}{\cancel{Aw}} + 3\overset{0}{\cancel{Az}}$$

$$\Rightarrow Av = c$$

This prove v must be the solution of $Ax = c$ ※

Another discussion, find the solution of $Ax = c$ by v and w vectors

For w, $Aw = c$. For z, $A(k \cdot z) = k(A \cdot v) = 0$, $k \in \mathbb{R}$

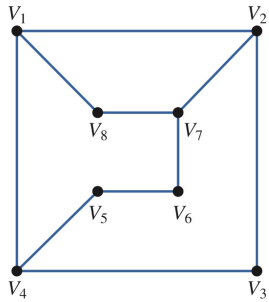
combine two terms $\Rightarrow Aw + A \cdot (kz) = c + 0$

$$\Rightarrow A(w + kz) = c$$

So all possible solutions $\Rightarrow x = w + kz$, $k \in \mathbb{R}$

6.

6. Consider the graph



(a) Determine the adjacency matrix A for the graph and enter it in MATLAB.

$$A = \begin{array}{c|cccccccc} & V_1 & V_2 & V_3 & V_4 & V_5 & V_6 & V_7 & V_8 \\ \hline V_1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ V_2 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ V_3 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ V_4 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ V_5 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ V_6 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ V_7 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ V_8 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{array}$$

(a)

```
C1 = [0 1 0 1 0 0 0 1]';
C2 = [1 0 1 0 0 0 1 0]';
C3 = [0 1 0 1 0 0 0 0]';
C4 = [1 0 1 0 1 0 0 0]';
C5 = [0 0 0 1 0 1 0 0]';
C6 = [0 0 0 0 1 0 1 0]';
C7 = [0 1 0 0 0 1 0 1]';
C8 = [1 0 0 0 0 0 1 0]';
A = [C1 C2 C3 C4 C5 C6 C7 C8];
```

(b)

```
>> HW2_6
A2:
```

	1	2	3	4	5	6	7	8
V_1	3	0	2	0	1	0	2	0
V_2	0	3	0	2	0	1	0	2
V_3	2	0	2	0	1	0	1	0
V_4	0	2	0	3	0	1	0	1
V_5	1	0	1	0	2	0	1	0
V_6	0	1	0	1	0	2	0	1
V_7	2	0	1	0	1	0	3	0
V_8	0	2	0	1	0	1	0	2

(i) $V_1 \to V_7$, 2 walks, (ii) $V_4 \to V_8$, 1 walk

(iii) $V_5 \to V_6$, 0 walk, (iv) $V_8 \to V_3$, 0 walk

(c)

A4:

18	0	13	0	9	0	15	0
0	18	0	15	0	9	0	13
13	0	10	0	7	0	10	0
0	15	0	15	0	8	0	10
9	0	7	0	7	0	8	0
0	9	0	8	0	7	0	7
15	0	10	0	8	0	15	0
0	13	0	10	0	7	0	10

A6:

119	0	86	0	64	0	103	0
0	119	0	103	0	64	0	86
86	0	63	0	47	0	73	0
0	103	0	93	0	56	0	73
64	0	47	0	38	0	56	0
0	64	0	56	0	38	0	47
103	0	73	0	56	0	93	0
0	86	0	73	0	47	0	63

A8:

	1	2	3	4	5	6	7	8
1	799	0	577	0	436	0	697	0
2	0	799	0	697	0	436	0	577
3	577	0	418	0	316	0	501	0
4	0	697	0	614	0	381	0	501
5	436	0	316	0	243	0	381	0
6	0	436	0	381	0	243	0	316
7	697	0	501	0	381	0	614	0
8	0	577	0	501	0	316	0	418

For A^4

$V_1 \to V_7$: 15 $\quad V_4 \to V_8$: 10

$V_5 \to V_6$: 0 $\quad V_8 \to V_3$: 0

For A^6

$V_1 \to V_7$: 103 $\quad V_4 \to V_8$: 73

$V_5 \to V_6$: 0 $\quad V_8 \to V_3$: 0

For A^8

$V_1 \to V_7$: 697 $\quad V_4 \to V_8$: 501

$V_5 \to V_6$: 0 $\quad V_8 \to V_3$: 0

$(1 \le i \le 8)$ $(1 \le j \le 8)$

from the result, we can assume from V_i and V_j, when $i+j$ is odd, then there would be no walks of even length. ※

(d)

A3:

	1							
1	0	7	0	6	0	3	0	5
2	7	0	5	0	3	0	6	0
	0	5	0	5	0	2	0	3
	6	0	5	0	4	0	4	0
	0	3	0	4	0	3	0	2
	3	0	2	0	3	0	4	0
	0	6	0	4	0	4	0	5
	5	0	3	0	2	0	5	0

A5:

0	46	0	40	0	24	0	33
46	0	33	0	24	0	40	0
0	33	0	30	0	17	0	23
40	0	30	0	23	0	33	0
0	24	0	23	0	15	0	17
24	0	17	0	15	0	23	0
0	40	0	33	0	23	0	30
33	0	23	0	17	0	30	0

A7:

0	308	0	269	0	167	0	222
308	0	222	0	167	0	269	0
0	222	0	196	0	120	0	159
269	0	196	0	149	0	232	0
0	167	0	149	0	94	0	120
167	0	120	0	94	0	149	0
0	269	0	232	0	149	0	196
222	0	159	0	120	0	196	0

For A^3

$V_1 \to V_7$: 0 $\quad V_4 \to V_8$: 0

$V_5 \to V_6$: 3 $\quad V_8 \to V_3$: 3

For A^5

$V_1 \to V_7$: 0 $\quad V_4 \to V_8$: 0

$V_5 \to V_6$: 15 $\quad V_8 \to V_3$: 23

For A^7

$V_1 \to V_7$: 0 $\quad V_4 \to V_8$: 0

$V_5 \to V_6$: 94 $\quad V_8 \to V_3$: 159

the conjecture in part (c) failed in odd length. As we pick randomly V_i and V_j for $i+j$ is odd, then we find that the number of walks is nonzero.
$(1 \le i \le 8)$ $(1 \le j \le 8)$
In opposite, V_i and V_j has no walks when $i+j$ is even ※

By easy induction,

① when length k is odd, there would be any walks when $i+j$ is odd, so there could be walks from V_i to V_j for odd k length as $i+j+k$ is even

② when length k is even, there would be any walks when $i+j$ is even so there could be walks from V_i to V_j for even k length as $i+j+k$ is even

①+② ⇒ In summary, there exist walks of any length $k \in \mathbb{Z}$ from V_i to V_j when $i+j+k$ is even. ※